

Essentials of Artificial Intelligence



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Table of Contents:

1.	Introduction	3
2.	Problem Statement.....	3
3.	Project Objectives	3
4.	Dataset Description	4
5.	Tools and Technologies Used	5
6.	Data Understanding and Analysis	6
7.	Feature Engineering	7
8.	Exploratory Data Analysis (EDA)	8
9.	Machine Learning Workflow	10
10.	Machine Learning Models	11
11.	Model Comparison.....	12
12.	Model Evaluation	12
13.	Streamlit Application Development	13
14.	Cyber Risk Assessment.....	15
15.	Limitations.....	16
16.	Future Work	16
17.	Conclusion	16

Table of Figures:

Figure 1: Dataset shape.....	4
Figure 2: Sample Dataset (First Five Rows)	5
Figure 3: Dataset Information	6
Figure 4: Missing Values.....	6
Figure 5: Duplicates Urls	6
Figure 6: Class Distribution Graph	7
Figure 7: Feature Dataset.....	8
Figure 8: Feature Shape	8
Figure 9: Correlation Heatmap.....	9
Figure 10: Feature Importance Graph.....	10
Figure 11: Target Classes	10
Figure 12: Train-Test Split.....	11
Figure 13: Logistic Regression Accuracy.....	11
Figure 14: Decision Tree Accuracy	11
Figure 15: Random Forest Accuracy.....	11
Figure 16: Model Comparison.....	12
Figure 17: Classification Report	12
Figure 18: Confusion Matrix Graph.....	13
Figure 19: Intelligent Agent URL Detection Dashboard	14
Figure 20: Benign URL Detection	14
Figure 21: Phishing URL Detection.....	15
Figure 22: Analysis Report Download	15

LAB PROJECT

Intelligent Agent-Based Malicious URL Detection and Cyber Risk Assessment System

1. Introduction

Cybersecurity threats continue to increase due to the rapid growth of internet usage. Among these threats, malicious URLs are commonly used to distribute malware, perform phishing attacks, and compromise user information. Detecting such URLs manually is difficult because attackers continuously create new malicious websites that appear legitimate.

The objective of this project is to develop an Intelligent Agent-Based Malicious URL Detection and Cyber Risk Assessment System using Machine Learning techniques. The system analyzes URL characteristics and predicts whether a URL is benign, phishing, malware, or defacement. The project also provides a risk assessment mechanism and security recommendations to assist users in making informed decisions before visiting suspicious websites.

2. Problem Statement

Cybercriminals frequently use malicious URLs to steal credentials, distribute malware, and perform various cyberattacks. Traditional blacklist-based approaches cannot detect newly generated malicious URLs effectively.

Therefore, an intelligent machine learning-based solution is required to automatically analyze URL characteristics and classify URLs into different security categories while providing risk assessment and security recommendations.

3. Project Objectives

The objectives of this project are:

- Detect malicious URLs using machine learning algorithms.
- Classify URLs into benign, phishing, malware, and defacement categories.
- Perform cyber risk assessment based on prediction confidence.

- Compare multiple machine learning algorithms.
- Develop a user-friendly Streamlit web application.
- Generate downloadable security reports.

4. Dataset Description

Dataset Information

Dataset Name: Malicious Phish Dataset

Source: Kaggle

Dataset Characteristics

Item	Value
Dataset	Malicious Phish Dataset
Total Records	651,191
Sample Used	50,000
Features	URL
Target Variable	Type

Classes

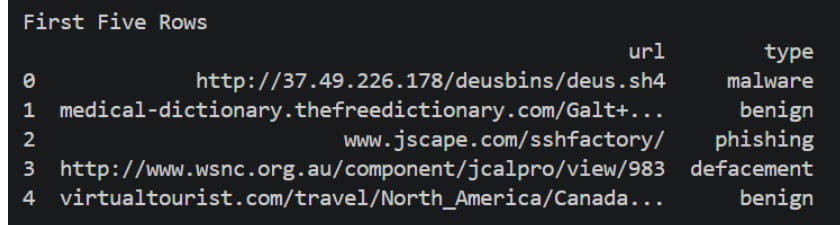
- Benign
- Phishing
- Malware
- Defacement

Dataset Shape

```
Original Shape  
(651191, 2)  
  
Sample Shape  
(50000, 2)
```

Figure 1: Dataset shape

Sample Dataset



	url	type
0	http://37.49.226.178/deusbins/deus.sh4	malware
1	medical-dictionary.thefreedictionary.com/Galt+...	benign
2	www.jscape.com/sshfactory/	phishing
3	http://www.wsnc.org.au/component/jcalpro/view/983	defacement
4	virtualtourist.com/travel/North_America/Canada...	benign

Figure 2: Sample Dataset (First Five Rows)

5. Tools and Technologies Used

Programming Language

- Python

Development Environment

- Visual Studio Code

Libraries

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn
- Joblib
- Streamlit

Machine Learning Algorithms

- Logistic Regression
- Decision Tree
- Random Forest

6. Data Understanding and Analysis

The dataset was analyzed to understand its structure, identify missing values, duplicate records, and class distribution.

Dataset Information

```
Dataset Info
<class 'pandas.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 2 columns):
 #   Column  Non-Null Count  Dtype  
---  -
 0    url     50000 non-null   str    
 1    type    50000 non-null   str    
dtypes: str(2)
memory usage: 4.0 MB
None
```

Figure 3: Dataset Information

Missing Values Analysis

The dataset was checked for missing values.

Result: No missing values were found.

```
Missing Values
url      0
type     0
dtype: int64
```

Figure 4: Missing Values

Duplicate Records

Duplicate URLs were identified.

Result: 316 duplicate records were detected.

```
Duplicates
316
```

Figure 5: Duplicates Urls

Class Distribution

The dataset contains four URL categories:

- Benign
- Defacement

- Malware
- Phishing

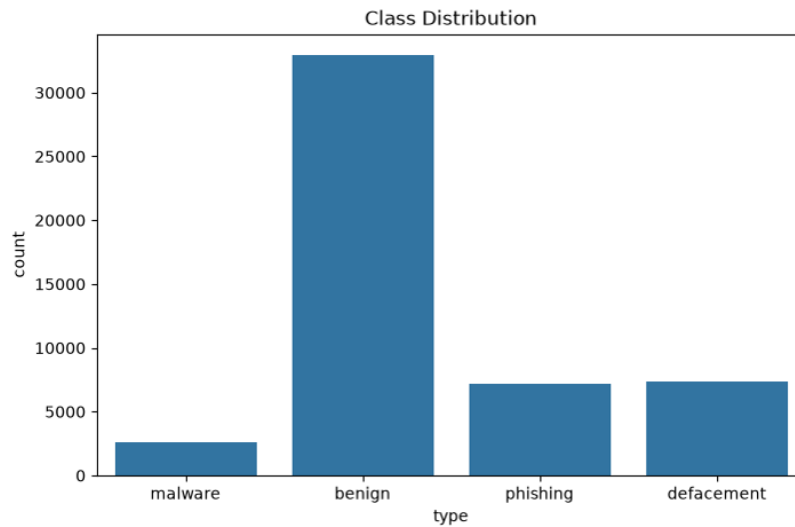


Figure 6: Class Distribution Graph

7. Feature Engineering

Machine learning models cannot directly understand URL strings. Therefore, multiple numerical features were extracted from each URL.

Extracted Features

Feature	Description
URL Length	Total URL length
Domain Length	Length of domain name
Number of Dots	Count of '.'
Number of Hyphens	Count of '-'
Number of Slashes	Count of '/'
Number of Digits	Numeric character count
HTTPS	Presence of HTTPS protocol

Number of Subdomains	Number of subdomains
Special Characters	Count of special symbols
IP Address Detection	Presence of IP address
Suspicious Words	Security-related keywords
Entropy	URL randomness measurement

Feature Dataset

```

Feature Dataset
url_length  domain_length  num_dots  num_hyphens  num_slashes  ...  num_subdomains  special_chars  has_ip  suspicious_words  entropy
0          38             13         4           0           4  ...           2           0         1           0  4.195296
1          61             40         2           1           3  ...           1           1         0           0  4.152943
2          33             14         2           0           4  ...           1           0         0           0  3.922590
3          49             15         3           0           6  ...           2           0         0           0  4.290345
4         128             18         2           5           8  ...           0           3         0           0  4.722947
[5 rows x 12 columns]

```

Figure 7: Feature Dataset

Feature Shape

```

Feature Shape
(50000, 12)
type
benign      32914
defacement  7383
phishing    7144
malware     2559
Name: count, dtype: int64

```

Figure 8: Feature Shape

8. Exploratory Data Analysis (EDA)

EDA was performed to understand relationships among extracted features.

Correlation Heatmap

The correlation heatmap shows relationships between URL features and helps identify feature importance.

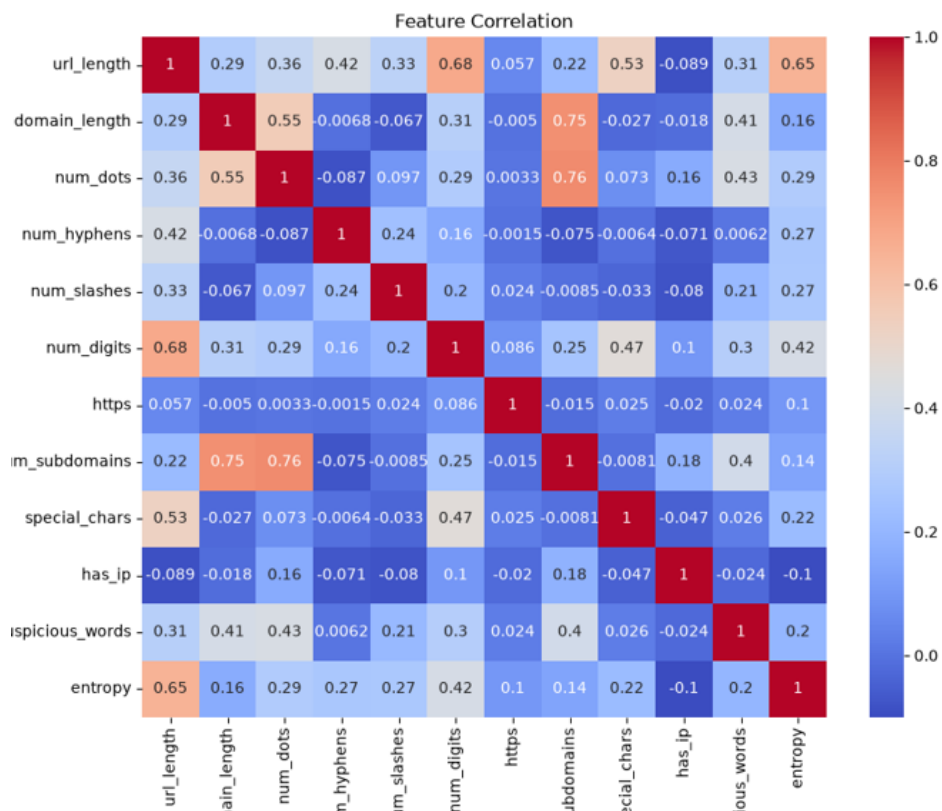


Figure 9: Correlation Heatmap

Feature Importance Analysis

Random Forest feature importance was used to identify the most influential features.

The most important features were:

- Number of Slashes
- Entropy
- URL Length
- Number of Dots
- Domain Length

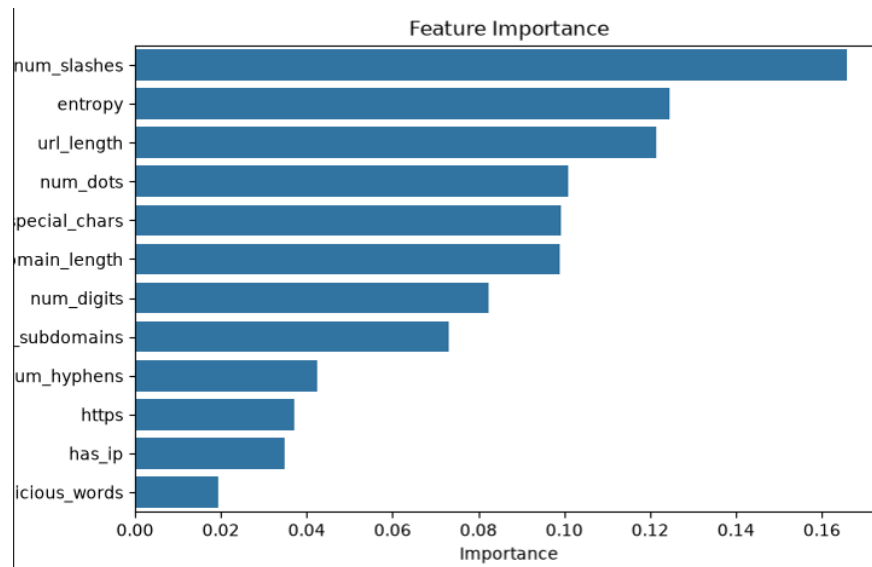


Figure 10: Feature Importance Graph

9. Machine Learning Workflow

- *Feature Matrix and Target Variable*

Input Features (X): Extracted URL Features

Target Variable (y): URL Category

- *Label Encoding*

The target classes were converted into numerical format using Label Encoding.

```
Classes
['benign' 'defacement' 'malware' 'phishing']
```

Figure 11: Target Classes

Train-Test Split

The dataset was divided into:

- Training Data = 80%
- Testing Data = 20%

Random State = 42

Stratified Sampling = Enabled

```
Train Shape
(40000, 12)

Test Shape
(10000, 12)
```

Figure 12: Train-Test Split

10. Machine Learning Models

Three machine learning models were implemented and compared.

- *Logistic Regression*

Logistic Regression was used as the baseline classification model.

```
LOGISTIC REGRESSION
Accuracy: 0.7372
```

Figure 13: Logistic Regression Accuracy

Accuracy: 73.72%

- *Decision Tree Classifier*

Decision Tree was used to improve classification performance.

```
DECISION TREE
Accuracy: 0.8210
```

Figure 14: Decision Tree Accuracy

Accuracy: 82.10%

- *Random Forest Classifier*

Random Forest was selected as the final model due to its superior performance.

```
RANDOM FOREST
Accuracy: 0.8744
```

Figure 15: Random Forest Accuracy

Accuracy: 87.44%

11. Model Comparison

Model Comparison Table

Model Comparison		
	Model	Accuracy
0	Logistic Regression	0.7372
1	Decision Tree	0.8210
2	Random Forest	0.8744

Figure 16: Model Comparison

Best Model Selection

Random Forest achieved the highest accuracy and was selected as the final model.

12. Model Evaluation

Classification Report

The classification report provides precision, recall, and F1-score for each class.

Classification Report				
	precision	recall	f1-score	support
benign	0.89	0.96	0.92	6583
defacement	0.86	0.74	0.79	1476
malware	0.96	0.83	0.89	512
phishing	0.77	0.65	0.71	1429
accuracy			0.87	10000
macro avg	0.87	0.79	0.83	10000
weighted avg	0.87	0.87	0.87	10000

Figure 17: Classification Report

Confusion Matrix

The confusion matrix illustrates the classification performance across all categories.

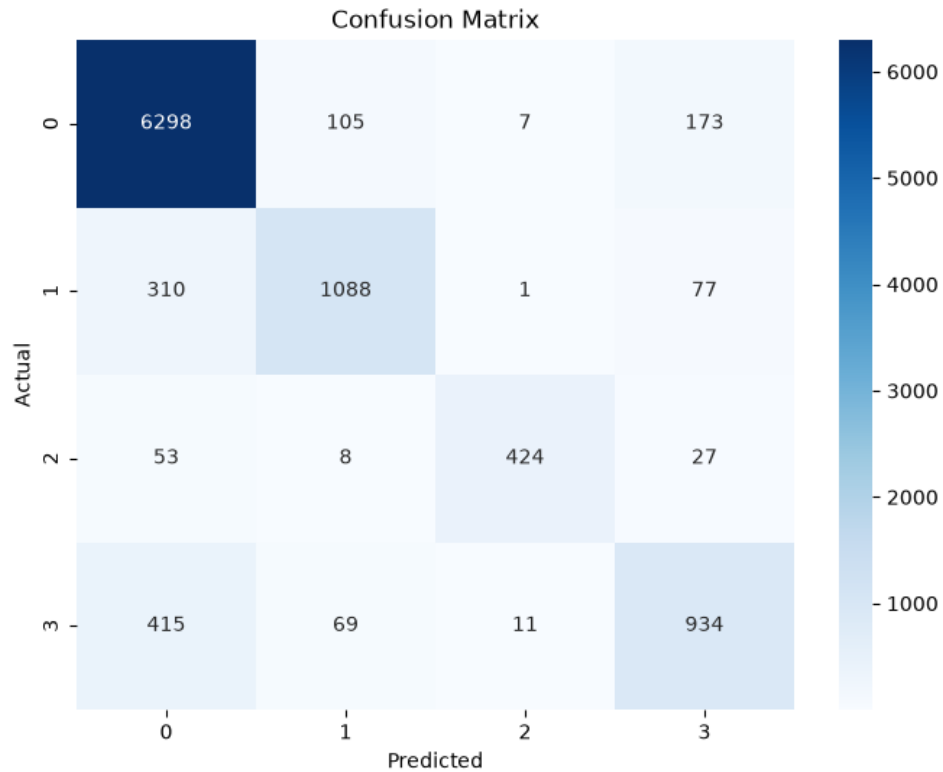


Figure 18: Confusion Matrix Graph

13. Streamlit Application Development

A web-based application was developed using Streamlit.

Application Features

- URL Input Interface
- Machine Learning Prediction
- Risk Assessment
- Security Recommendations
- Probability Visualization
- Downloadable CSV Report

Intelligent Agent Based Malicious URL Detection System

Enter a URL below and the AI model will analyze whether it is benign, phishing, malware, or defacement.

Enter URL

|

Analyze URL

Intelligent Agent Based Malicious URL Detection and Cyber Risk Assessment System

Figure 19: Intelligent Agent URL Detection Dashboard

Benign URL Detection

Example:

google.com

Enter URL

https://google.com

Analyze URL

Prediction: benign

Risk Level: LOW

Confidence: 99.00%

Recommendation

Safe website detected. Continue browsing normally.

Figure 20: Benign URL Detection

Phishing URL Detection

Example:

paypal-login-secure-update.com

Enter URL

paypal-login-secure-update.com

Analyze URL

Prediction: phishing

Risk Level: CRITICAL

Confidence: 99.67%

Recommendation

Do NOT enter passwords. Do NOT enter banking details. Leave website immediately.

Figure 21: Phishing URL Detection

Download Report Functionality

Analysis Report

	URL	Prediction	Risk Level	Confidence
0	http://paypal-login-secure-update.com	phishing	CRITICAL	99.6667

Download Report

Figure 22: Analysis Report Download

14. Cyber Risk Assessment

The system calculates risk levels based on model confidence.

Risk Percentage	Level
0–29%	Low

30–59%	Medium
60–79%	High
80–100%	Critical

The generated risk level assists users in understanding the potential threat associated with a URL.

15. Limitations

The current system primarily relies on handcrafted statistical URL features such as URL length, domain length, entropy, number of dots, and suspicious keywords.

Although the model achieves good classification performance, some legitimate websites may occasionally be misclassified because the model does not fully analyze the semantic meaning of domain names.

Therefore, additional verification mechanisms may be required for improved reliability.

16. Future Work

Future enhancements may include:

- TF-IDF based URL text analysis
- Deep Learning Models
- Real-time Threat Intelligence Integration
- Domain Reputation Services
- Browser Extension Integration
- Automated Threat Reporting

These improvements can further enhance detection accuracy and reduce false positives.

17. Conclusion

This project successfully developed an Intelligent Agent-Based Malicious URL Detection and Cyber Risk Assessment System using machine learning techniques. The dataset was analyzed, processed, and transformed into meaningful numerical features. Logistic Regression, Decision

Tree, and Random Forest algorithms were implemented and evaluated. Among all models, Random Forest achieved the highest accuracy of 87.44% and was selected as the final model.

A Streamlit-based web application was developed to provide real-time URL analysis, cyber risk assessment, security recommendations, and downloadable reports. The results demonstrate the effectiveness of machine learning techniques in identifying malicious URLs and assisting users in making safer browsing decisions.

References

1. Malicious Phish Dataset (Kaggle)
2. Pandas Documentation
3. NumPy Documentation
4. Matplotlib Documentation
5. Seaborn Documentation
6. Scikit-Learn Documentation
7. Streamlit Documentation
8. Joblib Documentation

Appendix A – Complete Python Code

(Attach url_detection.py)

Appendix B – Streamlit Application Code

(Attach app.py separately)